Executive Summary: Data Professional Job Market

1. Project Objective and Dataset Scope

The objective of this project is to analyze the job market landscape for data professionals, focusing on demand, compensation, and work modalities. The dataset encompasses a total of **2,076 job postings** categorized into three main roles: Data Analytics, Data Engineering, and Data Science. The geographic scope is primarily concentrated in the United States, with job postings tracked predominantly across the months of March and April.

2. Key Findings

- **Role Demand:** Data Analytics is the most highly sought-after role with **839 postings**, followed by Data Engineering (701) and Data Science (536).
- **Compensation Variations:** Data Science commands the highest compensation with a median annual salary of **$170,000**. Data Engineering follows at $142,500, while Data Analytics has a median salary of $110,000. The overall median salary across all data roles is **$135,200**.
- **Work Modality:** Despite industry shifts toward flexible work, fully remote jobs make up only a small fraction of this specific market, representing just **4.43%** of the total postings.

3. Practical Recommendations

- **For Job Seekers and Diploma Students:** With Data Analytics showing the highest volume of available postings (839 jobs), entry-level candidates should prioritize building strong foundational analytics portfolios to target this broad opportunity base.
- **For Career Advancement:** Professionals aiming for higher compensation should consider specializing in Data Science or Data Engineering. These highly technical fields offer significantly higher median salaries ($170,000 and $142,500, respectively) compared to general analytics roles.

4. Data Limitations

- **Salary Coverage:** A major limitation is that only **31.45%** of the job postings (653 out of 2,076) explicitly included salary information. Data Engineering had the lowest visibility with only 28.82% coverage, meaning the calculated trends might not perfectly capture the entire market.

- **Geographical Bias:** The geographical data is heavily skewed towards the United States (with top locations including New York, Austin, and Atlanta), which restricts the ability to apply these insights to global job markets.

5. Handling of Missing Salary Values

> Given the low salary coverage (31.45%), missing salary values were strictly excluded from aggregate calculations rather than being imputed or replaced with zeroes. The reported median annual salary metrics are based exclusively on the 653 postings where compensation data was provided. This method ensures that the median figures accurately reflect verified market offers without being artificially lowered or skewed.